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SUPERSEDING
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MILITARY SPECIFICATION

DOUGH PROOFING BOXES, STEAM HEATED (STEAM, OR STEAM AND WATER, HUMIDIFIED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of dough proofing box.

1.2 Classification. The dough proofing boxes shall be the following sizes as specified (see 6.2):

Size	For use with garrison bakery Size No.	For use with central pastry kitchen Size No.
1	Bak-A-M	BA-6
2	Bak-B-M	BA-3, BA-5
3	Bak-C-M	----
5	Bak-G-M	BA-7
6	----	----
7 (see 6.1)	----	----

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Non-leaded: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-C-576 - Copper Flat Products With Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges, (Plate, Bar, Sheet, and Strip).
RR-P-58 - Pans, Bread.

FSC 7320

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MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-B-2316	-	Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery of.
MIL-R-2345	-	Rack, Bread, Bakers': 10 Shelf, Knockdown Type.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MS 20615	-	Rivet, Solid-Universal Head, Brass, Copper and Nickel-Copper Alloy.

LAWS AND REGULATIONS:

29 CFR 1910 - Occupational Safety and Health Act (OSHA) Standards

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprint of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Baking Industry Sanitation Standard Committee (BISSC)

Sanitation Standard No. 10

(Application for copies should be addressed to the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, N.Y. 10017.)

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American Society For Testing and Materials (ASTM)

ASTM Designation A167-70 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
ASTM Designation A176-71 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip.
ASTM Designation B117-64 - Method of Salt Spray (Fog) Testing.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Underwriters' Laboratories, Inc. (UL)

Standard No. 73 - Motor-Operated Appliances

(Application for copies should be addressed to the Underwriters' Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062; 1285 Walt Whitman Rd., Melville, L.I., NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups, and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.2 and 6.5).

* 3.2 Standard product. The dough proofing boxes delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with this specification.

3.3 Codes and standards. The dough proofing boxes shall comply with the applicable requirements of UL Standard No. 73, BISSC Standard No. 10, and OSHA Standards 29 CFR 1910, promulgated under the Occupational Safety and Health Act.

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the dough proofing boxes he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 73, BISSC Standard No. 10, and OSHA Standard 29 CFR 1910 as follows:

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3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the dough proofing boxes have been tested and conform to UL Standard No. 73.

3.3.1.2 BISSC. Acceptable evidence of meeting the requirements of BISSC shall be:

(1) A photostatic copy of the current BISSC certificate of authorization for the dough proofing boxes, or
(2) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the dough proofing boxes are being procured, indicating that the units have been tested and conform to the applicable requirements of BISSC Standard No. 10.

3.3.1.3 OSHA. Acceptable evidence of meeting the applicable safety and health standards of OSHA shall be a certificate of compliance, certifying that the dough proofing boxes furnished conform to OSHA Standards 29 CFR 1910.

3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial dough proofing boxes provided the completed item complies with all provisions of this specification.

3.4.1 Brass sheet. Brass sheet shall conform to QQ-B-613.

3.4.2 Copper.

3.4.2.1 Copper sheet. Copper sheet shall conform to QQ-C-576.

3.4.2.2 Copper rivets. Copper rivets shall conform to MS 20615.

3.4.3 Stainless steel. Stainless steel sheet shall conform to any type steel in the 300 or 400 series of ASTM A167 or A176, respectively.

3.5 Design. The dough proofing box shall consist of a paneled steel frame covered with material that will protect against corrosion and equipped with steam humidification. Doors shall be provided at two opposite ends, except sizes 5, 6, and 7 shall have doors at one end only. Rack accommodation shall be as specified in table I.

3.5.1 Air conditioning and humidification. The dough proofing box shall be equipped with an air processing unit for recirculating air in sufficient quantities to maintain a dry bulb temperature range of 90 Fahrenheit (F.) to 110 F. and a relative humidity range of 80 to 90 percent within the box. Unless otherwise specified (see 6.2), outside design temperatures shall be 70 F. dry bulb, 53 F. wet bulb. The unit shall be designed to proof a full load of 20-ounce loaves with the quantity of racks shown in table I in 50 minutes.

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TABLE I. Dough proofing box requirements

Dough proofing box size 1/	No. of bread racks per proofing box 2/	No. of 20 oz loaves per proofing box using bread pans 3/
1	4	800
2	6	1200
3	10	2000
5	3	600
6	2	400
7	2	320

1/ See 6.3.

2/ Racks conforming to MIL-R-2345.

3/ Bread pans in accordance with type II, style 3, size 3 of RR-P-58.

Each box shall be equipped with an automatic control system that is adjustable to selected points within the specified operating range and shall be capable of maintaining the air processed in the proofing room with a maximum dry bulb temperature differential of plus or minus 2 F., and a maximum relative humidity differential of not more than plus or minus 2-1/2 percent on either side of the point to which the controls are set. Manufacturer's test data shall prove to the satisfaction of the contracting officer that the assembled box and air processing unit meet the requirements of this specification.

3.5.2 Rack arrangement. The proofing box shall be designed to accommodate the number of bread racks specified in table I, in aisles of 2 racks per aisle except for sizes 5, 6, and 7, which shall accommodate one rack per aisle.

3.6 Construction.

3.6.1 Protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wetted in service shall be protected against electrolytic or galvanic corrosion by proper selection of materials, plating, isolation, insulation, paint area relationship, gaskets, or other means. Fasteners and fittings shall be of the same materials as the materials being joined or supported.

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3.6.2 Box.

3.6.2.1 Frame. The frame shall be constructed of structural steel shapes and shall be self-supporting before panels are applied. All steel used in the frame shall be hot dip galvanized to conform with requirements of 3.6.4. When galvanizing is removed by fabrication, it shall be restored to its original galvanized condition.

3.6.2.2 Panels. All panels shall be fabricated of at least the equivalent thermal conductivity of 1-inch of insulating board, having a maximum K-factor of 0.36 Btu per hour per square foot per 1-inch thickness per degree F. temperature difference. The inner panel surface shall be constructed from nominal 0.025 inch stainless steel in the 300 or 400 series with a 2B, or better, finish. The outer panel surface shall be steel sheet not less than 0.020 inch thick and shall be either galvanized to conform with requirements of 3.6.4 or covered with a standard paint to conform with requirements of 3.6.4 and then finished in white baked synthetic enamel. The covering sheets and insulating board shall be bonded together with waterproof cement in accordance with the manufacturers' commercial practice. The insulating material shall be of nonmoisture absorbing material such as foamed rubber, foamed plastics, or dried pressed fibre insulation board rendered impervious to moisture. The panels shall be butted together with a mastic sealer, molded rubber, or plastic seal to be air and moisture tight.

3.6.2.3 Doors. Doors shall be not less than 1-inch thick and designed in accordance with the manufacturers' standard product. Provision shall be made to prevent moisture from entering the insulation. The door surfaces shall be constructed from the same materials specified for the panels in 3.6.2.2. Suitably located metal guards or bumper plates shall be provided on both sides and edges of the doors to protect door panels from damage by racks entering or leaving the proofing box. Doors shall be hung with hinges designed and mounted to insure permanent alignment and shall be provided with latches or catches to insure tight closing. Means shall be provided for opening doors from the inside. All hardware, including hinges, exterior bumper bars, and latches shall be finished as specified in 3.8.

3.6.2.4 Door opening. Clear openings for rack entrance shall be not less than 34-1/2 inches wide by 78 inches high for sizes 1 to 6, and not less than 34-inches by 76-inches for size 7.

3.6.2.5 Rack guide rails. The dough proofing box shall be provided with one rack guide rail on each interior sidewall and other guide rails spaced between 30-1/2 and 31-1/2 inches apart between rails. The guide rails shall be the only means for guiding the rack through the box. The guide rails shall be 1/4-inch thick plate or equivalent angle or pipe construction supported at each end and at necessary intermediate points above the floor. The height of the guide rails shall be between 8 and 9 inches.

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3.6.2.6 Guard rails. Guard rails, consisting of 1-1/2 inch galvanized pipe rail fittings, and 3-inch channel sections shall be provided at the outside corners and side walls to protect the box from contact with racks, pan, trucks, and other rolling equipment.

3.6.2.7 Electrical requirements. Unless otherwise specified (see 6.2), all items shall be designed for operation on a nominal 208-volt, 60 hertz (Hz) 3-phase system. Motors or control system of 1/3 horsepower (hp) or less may be supplied for operation on a nominal 120-volt, 60 hertz (Hz), single phase (a.c.) system. Thermal overload protection shall be required on all motors.

3.6.2.8 Lighting. A sufficient number of lights shall be installed within each box to provide a general lighting level of not less than 20 foot candles when tested as specified in 4.4.4.

3.6.2.9 Inside clear height. The inside clear height of the proofing boxes shall be not less than 78-inches for sizes 1 to 6 and 76-inches for size 7. Each box shall be of sufficient size to accommodate the number of racks specified in table I, each rack being approximately 70-inches long, 29-inches wide and 72-inches high, except the rack used in size 7, shall be 43-1/4-inches long, 28-inches wide, and 70-1/2-inches high.

3.6.2.10 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.3).

3.6.3 Air processing equipment. The air processing unit shall consist of a conditioning chamber in which heat and moisture are supplied to the air under controlled conditions, and a centrifugal air mover. The air processing chamber and air mover or fan may be incorporated in a common unit casing or in separate units. In addition, an air cleaning system may be incorporated and shall be a direct water type or a recirculating water type with water pump. Air processing equipment shall meet the test specified in 4.4.1.

3.6.3.1 Air processing unit. The casing of each air processing unit and all parts coming in contact with free water shall be fabricated from metal that has been tested for corrosion-resistance as specified in 3.5.4. The casing shall be fabricated of copper sheet specified in 3.4.2.1; brass sheet specified in 3.4.1 not less than 0.027 inch thick; or cold rolled corrosion-resisting steel sheets specified in 3.4.3, not less than 0.030 inch thick. Casing shall be supported and stiffened by structural shapes, constructed of the same type material or material with a corrosion-resistant finish, welded, riveted in

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place with rivets specified in 3.4.2.2, or bolted in place with brass screws and nuts. Aluminum alloy sheet not less than 0.030 inch normal thickness also may be used. When aluminum is used, it shall be riveted in place with commercial brazier aluminum rivets and proper means shall be used to prevent electrolytic corrosion between aluminum and copper or brass parts. All water and steam pipe and fittings inside the conditioning chamber shall be brass or copper. External pipe and fittings, shall be of the material normally used by the manufacturer for this type of application.

3.6.3.2 Humidification. Humidification shall be accomplished either by direct steam injection into the airsteam within the unit or by steam injection in conjunction with a direct or recirculated water spray chamber. The design and capacity of the humidification section shall be such as to insure at least 90-percent saturation of the air leaving the unit. Water spray system shall be direct or shall be provided with a direct connected pump capable of handling the water required for the spray nozzles. A drip pan shall be provided to catch any leaks from the packing glands. Means shall be provided to prevent overflow of the drip pan and resultant damage by either an open or automatic float valve, actuated drain valve. Brass spray nozzles of the large orifice swirling nonclogging type which produce both an air cleaning and humidifying coarse spray shall be used.

3.6.3.3 Eliminators. Eliminators shall be provided when required for the proper functioning of the equipment furnished. The eliminators shall prevent passage of any particles of spray from the spray chamber.

3.6.3.4 Water sump. When a recirculating water spray system is provided, a watertight sump and an overflow pipe of sufficient capacity to prevent flooding shall be provided. A filter shall be provided over the pump suction intake to prevent passage of solid matter of a size that would clog the pump or damage the impeller. A float valve with seamless copper ball or plastic-float shall be provided for maintaining the water in the sump. The fresh water inlet shall be above the maximum overflow level by at least 1-1/2 times the diameter of the inlet line to prevent siphonage.

3.6.3.5 Heating. Each heating coil shall be of the extended-surface type constructed of copper coated steel or copper tube and shall be provided with fins with cast iron headers, or similar approved construction. The trap controlling the air heater may be drained into the spray washer sump when the addition of condensate will not affect the control of air processing. All heating equipment shall be designed for operation with a steam pressure of 10 to 15 pounds per square inch gage (psig). All external pipe and fittings shall be of a material normally used by the manufacturer for this type of application.

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3.6.3.6 Fans. The fan shall be of the centrifugal, multiblade type, connected to a standard motor of sufficient size to operate the fan at the required capacity under the actual usage conditions of the system and under conditions of continuous operation. The blower wheel shall be constructed of noncorrosive material or material properly treated to conform to corrosion-resistant requirements specified in 3.6.4. The fan shall recirculate evenly throughout the box a volume of air equal to 90 cubic feet plus or minus 10 percent per minute per rack. Compliance with the volumetric requirements shall be determined when tested as specified in 4.4.2.

3.6.3.7 Air distribution. The air distributing system shall provide means for circulating the air evenly within the proofing box (see 4.4.2). All ductwork shall be constructed of galvanized steel or aluminum not less than 0.026 inch thick, reinforced to protect against damage. Distributing ducts shall be protected against corrosion on inner surfaces of ducts. Drop ducts shall be fabricated from not less than 0.0625-inch-thick aluminum sheet. The openings, for circulating air within the box shall be of such arrangement and size as to produce an even distribution of air throughout the box.

3.6.3.8 Control system. Each control system shall be fully automatic and designed for controlling the temperature of the air discharging into the box. The control system shall be coordinated to compensate for change in temperature and humidity to produce the desired design temperature and humidity condition. The thermal bulb for control of the dry-bulb and wet bulb temperature shall be located in a position in contact with the average condition of air within the proofing box and shall be completely protected from any influence other than that of the air processed.

3.6.3.9 Auxiliary manual control. Means shall be provided for manually operating and controlling dry- and wet-bulb temperatures in the event the automatic control system becomes inoperative.

3.6.3.10 Indicators. The equipment shall be provided with temperature, humidity, and steam indicator(s) or any other indicators necessary in the design of the proofing box. The indicator(s) shall be located to permit easy reading and shall provide for calibration and adjustment.

3.6.4 Corrosion-resistance. Material treated to resist corrosion shall not show evidence of corrosion after being subjected to the 48 hour salt spray test specified in 4.3.2.2.

3.6.5 Guards. All rotating, reciprocating, or projecting parts so located as to endanger any person that may come in close proximity thereto, shall be enclosed or guarded. All steam lines and appurtenances exceeding a temperature of 150 F. and so located as to endanger personnel or create a fire hazard shall be guarded or covered with insulating material.

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3.7 Lubrication. Provisions shall be made for the proper lubrication of all moving parts. Seals shall be provided where necessary to prevent leakage of lubricant. At time of delivery, the unit shall be completely lubricated with the manufacturer's standard lubricant.

3.7.1 Lubricating chart. The supplier shall attach to each dough proofing box a lubricating chart indicating all points on the equipment requiring lubrication, grade of lubricant, and time interval when lubrication should be accomplished.

3.8 Finish. Unless otherwise specified herein, proofing boxes shall be finished in accordance with the manufacturer's standard practice, except the color of painted finish shall be white enamel. Hardware, including hinges, exterior bumper bars, and latches shall be galvanized or otherwise suitably protected against corrosion to conform with requirements in 3.6.4.

3.9 Marking for identification. The dough proofing box shall be marked in accordance with MIL-STD-130 on a photo-sensitive anodized aluminum plate conforming to MIL-P-514. The plate shall be located on the upper right hand corner opposite the door opening side. When two door opening sides are furnished, the plate may be attached at either end.

3.10 Workmanship.

3.10.1 Fabrication. All sharp burrs, slivers, or splinters shall be removed.

3.10.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have burrs removed. Washers or lock washers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be drawn tight.

3.10.3 Welding. Welded joints shall be complete, uniform, sound, and thoroughly fused, with no pits, undercuts, cracks, burn-holes, or fractures. All scale and flux deposits shall be removed. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall develop strength adequate to the parts connected. Welding process shall be in accordance with a welding code acceptable to the bureau or service concerned.

3.10.4 Electrical assembly. Wiring shall be properly joined, not cut or abraded, not loose at terminal ends, properly enclosed in conduit where required, provided with adequate slack, and not stripped of required insulation.

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3.10.5 Solder. All surfaces to be soldered shall be thoroughly cleaned prior to soldering. Solder joints shall be smooth with no pin holes.

3.10.6 Metal forming. All metal forming shall be done with appropriate tools to insure uniformity of size and shape. Sheet metal or structural sections used in fabrication of the boxes shall be free of kinks and bends. Shearing and chipping shall be done accurately and cleanly and shall be neatly finished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article is required, it shall be examined for defects listed in table II for dimensions specified, and tested as specified in 4.4.1, 4.4.2, 4.4.4, and 4.4.3 when applicable. Failure to pass any of the tests, the presence of any defect listed in table II, or any dimension not within specified requirements shall be cause for rejection of the first article.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements specified in referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process inspection.

4.3.2.1 In-process examination. Examination shall be made for the following to establish that no deviation is made from requirements specified in 3.6.2.2:

(a) Application of paint grip to outer panel.
(b) Insulating material application.
(c) Panels are butted.
(d) Provision for making insulation waterproof.

When a deviation from requirements is noted, correction shall be made to items and process.

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4.3.2.2 Salt spray tests. Components shall be subjected to a 48 hour salt spray test in accordance with ASTM B117 to determine compliance with corrosion-resistance requirements of 3.6.2.1, 3.6.2.2, 3.6.3.1, 3.6.3.6, 3.6.4, and 3.8. The lot shall consist of all like components offered for inspection at one time. The sample unit shall be a single component. Whenever a deviation is noted, correction shall be made to items and process.

4.3.3 End item inspection. The lot shall consist of all the dough proofing boxes of one size offered for acceptance at one time. The sample unit for this inspection shall be one completely fabricated dough proofing box.

4.3.3.1 Visual examination. The completely assembled end item shall be examined for defects in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished in accordance with manufacturer's standard commercial product.	X	
	Galvanized coating removed by fabrication.	X	
	Blistered, peeled, chipped, or area of film (painted items).		X
	Painted finish not white.	X	
Construction and workmanship, general (applicable to all components and assemblies)	Part missing or not specified type.	X	
	Any component fractured, split, punctured, dented, creased, deteriorated, bowed, sprung, or malformed.		X
	Part loose.		X
	Part not in proper alignment.	X	
	Sharp burr, sliver, or edge.	X	
	Functioning component that is inoperative or does not operate as intended.	X	
	Adjustable assembly that cannot be adjusted.		X
	Adjustable assembly not properly adjusted to perform the intended function.	X	
	Component not readily accessible for servicing, where required.	X	
	Joint not protected against electrolytic or galvanic corrosion.	X	

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding and brazing (where required)	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused.	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed.		X
Soldering	Not adherent or incomplete, not clean (flux residue), not smooth, or pin holes in solder.		X
Threaded fasteners	Missing, broken, stripped, fractured, or loose.		X
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminals, or not inclosed in conduit (where required).	X	
	Adequate slack not provided to relieve strain or required insulation stripped from wiring.	X	
Gages and instruments	Not specified capacity or type.	X	
	Inoperative or otherwise defective affecting use (visual determination).	X	
	Not mounted correctly, but will properly function.		X
Lubrication (where applicable)	Not lubricated where required.	X	
Assembly	Component not properly assembled or secured.	X	
	Unit perceptibly out of square or alignment.	X	
Identification marking	Missing, incomplete, not legible, not specified type, not in proper location.		X

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4.3.3.2 Dimensional examination. Inspection shall be made of the end item for compliance with dimensions specified. Any dimension not within the specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item. Each box shall be tested in accordance with 4.4.1. When a first article is not required, the initial production unit shall be tested as specified in 4.4.1, 4.4.2 and 4.4.4, and 4.4.3 when applicable. At the option of the Government any or all tests may be requested if there is a change in design, material, construction or other departure subsequent to approval of the initial production unit. When the box is to be installed outside the Continental United States (see 6.2) each box shall either be tested as specified above, or the supplier shall furnish evidence that a box identical to that offered has successfully passed the tests specified in 4.4.1, 4.4.2, 4.4.4 and 4.4.3 when applicable, in the manufacturer's plant or in a commercial application. The Government reserves the right to check-test each box to determine the validity of the evidence produced.

4.3.4 Standard compliance. Proof of compliance with the BISSC, UL and OSHA requirements of 3.3 shall be submitted to the Government representative.

4.3.5 Examination of preparation for delivery. An examination shall be made to determine that preparation for delivery as specified in section 5 of this specification is in conformance with MIL-B-2316.

4.4 Tests.

4.4.1 Operation. Air processing unit controls shall be connected to a power supply as specified in the contract or order and a water supply to determine conformance with 3.6.3. Steam system shall be leak-tested with water at a minimum of 40 psi for 30 minutes. Any pressure drop is cause for rejection of the item.

4.4.2 Air recirculation. Evenness of air distribution shall be determined for compliance with 3.6.3.7 by measuring air velocities in a completely assembled box with air-processing system mounted on its permanent location but not necessarily connected to utilities except for electric power. The full load of bread racks (see table I), with bread pans in place shall be distributed in the box as they would be in actual service. Air velocity shall be measured in the geometric center of each aisle. Measurement shall be made by an appropriate instrument of best commercial accuracy. Measurements at any two of the above locations shall not differ by more than 20 percent of the higher reading within the limits of 3.6.3.6; a variation of more than 20 percent shall constitute failure of the test.

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4.4.3 Electromagnetic Compatibility Test. When electromagnetic compatibility is required, the first article or initial production unit, as applicable, shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the Contracting Officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

4.4.4 Foot candle test. The lighting level shall be measured using a Weston Illuminator Meterm model 756 or equivalent to determine compliance with 3.6.2.8. Nonconformance with 3.6.2.8 shall constitute failure of test.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Preservation, packaging, and packing and marking, for each level of protection shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use. The dough proofing boxes covered by this specification are used to proof dough in warm, moist air, in a uniform manner prior to baking. Size 7 is used with the following items:

MIL-R-43814; Rack, Mobile, Bakery Products and Meat; With Shelves
MIL-O-43816; Oven, Baking and Roasting, Roll-Through, Convection, Gas
MIL-O-43815; Oven, Baking and Roasting, Roll-Through, Convection, Electric

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size proofing box to be furnished (see 1.2).
(c) When a first article is required (see 3.1, 4.2, and 6.5).
(d) Outside design temperature if other than specified (see 3.5.1).
(e) When electrical characteristics are other than specified (see 3.6.2.7).
(f) When electromagnetic compatibility is required (see 3.6.2.10).
(g) When installation is outside Continental U.S. (see 4.3.3.3).
(h) Selection of the applicable levels of preservation, packaging and packing (see 5.1).
(i) When supplier is to furnish supervision of installation or demonstration service to items. If required, the location, approximate date, the person or office to report to, and any accommodation or service to be supplied by the ordering agency will be stated (see 6.4).

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6.2.1 Contract data requirements. Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423: Contract Data Requirements List, and cited in the contract.

6.3 Sizes of bakeries and pastry kitchens specified in 1.2 are in Table of Allowances 50-911, Department of the Army. The bakeries are covered in section III, part 7 for Garrison Bread Bakeries and the pastry kitchens are covered in part 8 for Central Pastry Kitchens.

6.4 Supervision. The following information should be included in procurement documents. When specified (see 6.2), supervision of installation shall be required of the supplier for dough proofing boxes to be erected in the Continental U.S.A. The supplier shall furnish the services of a competent superintendent who shall be familiar with the construction, erection, operation, maintenance and repair of the equipment and the required utility connections. Material required for complete installation other than specified herein, and all installation labor, will be furnished by others. All utilities will be extended and connected to the equipment by others. The superintendent shall supervise the installation of the equipment and all utility connections thereto, using the material as stated above. Supervision of utility connections shall include all work from points within 10 feet of the dough proofing box. Upon completion of the work, the superintendent shall turn the equipment over to the contracting officer or his representative in proper operating condition and shall instruct personnel designated by the contracting officer in the proper operation.

6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.6 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the

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Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Review activities:

Army - MD
Navy - MC, MS

User activity:

Navy - YD

Preparing activity:

Army - GL

Project No. 7320-0527

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-D-40167C Dough Proofing Boxes,Steam Heated(Steam,orSteam and Water,Humidified)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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